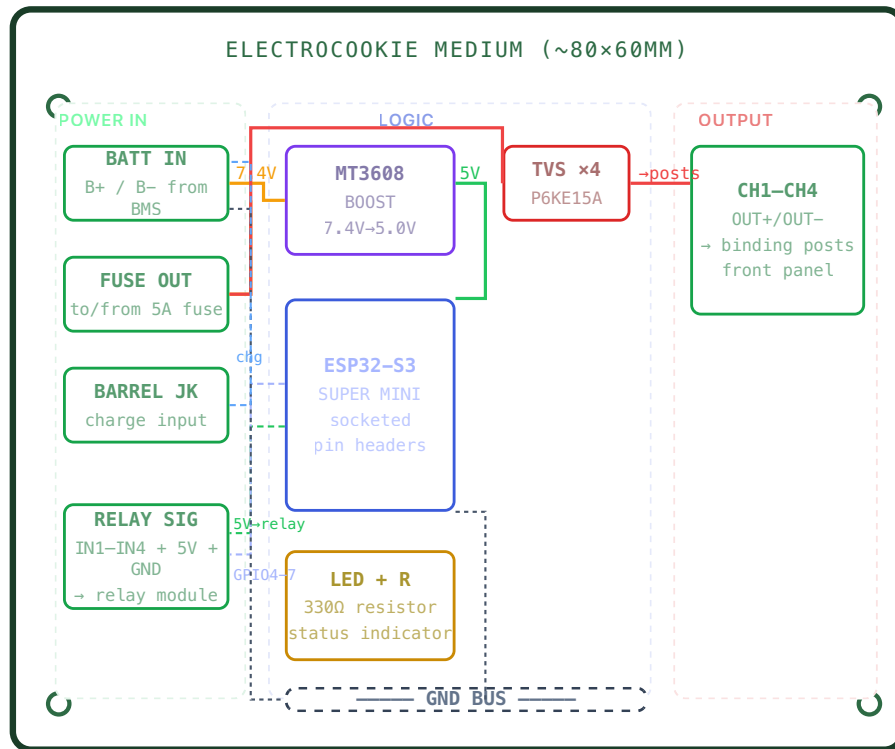


PCB Layout – Medium ElectroCookie

~80×60mm board · components to scale · relay module mounts separately



COMPONENT KEY

- U1** **ESP32-S3 Super Mini**
Socketed via 2× 8-pin female headers. Center of board for short GPIO runs to both relay signals and LED.
- U2** **MT3608 Boost**
7.4V → 5V. Pre-adjust output before soldering in. Position near battery input terminals.
- D1-4** **P6KE15A TVS Diodes**
One per channel. Solder inline on the output rail between relay NO terminals and binding post screw terminals.
- LED** **Status LED + 330Ω R**
GPIO → resistor → LED → GND. Solder directly to board, wire extends to front panel LED hole.
- J1-6** **Screw Terminals**
All external connections use screw terminals – battery, fuse, barrel jack, relay signals, output to binding posts. Makes wiring and troubleshooting clean.

TRACE COLORS

- 7.4V** Raw battery rail (switching)
- 5V** Boosted logic rail

OUT	Switched igniter output
SIG	GPIO signal lines
GND	Common ground bus

BUILD NOTES

1. Socket the ESP32-S3 with female pin headers so it's removable. The Super Mini is narrow — use two rows of 8-pin headers.
2. Mount MT3608 first, adjust to 5.0V with a multimeter before anything else is connected. Use a small flat screwdriver on the pot — it's sensitive.
3. The relay module mounts separately on the Hammond's internal bottom surface using M3 screws through its existing corner holes. Run a 6-wire ribbon (GND, 5V, IN1, IN2, IN3, IN4) from the relay signal screw terminals on the PCB to the relay board.
4. TVS diodes go on the output side of the relay NO contacts — between the relay module and the binding post terminals. Cathode toward the positive output.
5. The LED lead will need to be long enough to reach the front panel hole. Solder a 15cm wire extension and use heat shrink on the joints.
6. Keep 7.4V power traces (battery, fuse, output) on one side of the board and 5V/signal traces on the other where possible. Helps with noise isolation.
7. Use 18 AWG wire for all 7.4V power connections. 22 AWG is fine for 5V logic and GPIO signals.